GATE 2021 Computer Science and Information Technology

Instructions: Choose the correct option. Negative marking applies.

Q1. Which of the following page replacement algorithms suffers from Belady's Anomaly?

(A) Optimal Page Replacement
(B) LRU
(C) FIFO
(D) Both LRU and FIFO

Q2. Consider the following recurrence relation: T(n) = 2T(n/2) + O(n). What is the time complexity?

(A) O(n)
(B) O(n log n)
(C) O(n^2)
(D) O(log n)

Q3. In a standard Ethernet network, what is the minimum frame size?

(A) 32 bytes
(B) 64 bytes
(C) 128 bytes
(D) 256 bytes

Q4. Let R(A, B, C, D) be a relational schema with functional dependencies: A -> B, B -> C, C -> D. Which normal form is R in?

(A) 1NF only
(B) 2NF only
(C) 3NF but not BCNF
(D) BCNF